

Heatwave Survival Guide — Heat Exhaustion and Heat Stroke First Aid

This guide covers how to protect yourself and others during a heatwave, how to recognise the stages of heat illness, and how to treat heat exhaustion and heat stroke. Based on CDC, Mayo Clinic, WHO, and NDMA guidelines. India experiences severe heatwaves annually across Rajasthan, Gujarat, Telangana, Andhra Pradesh, Odisha, and the Indo-Gangetic plain. Heat kills more people in India than any other weather event.

HEAT STROKE IS A LIFE-THREATENING EMERGENCY. It kills within minutes if not treated. Call emergency services immediately AND begin cooling the person at the same time. Do not wait for help to arrive before cooling.

SECTION 1: THREE STAGES OF HEAT ILLNESS

Heat illness progresses through stages. Recognising the early stage allows you to stop it before it becomes life-threatening.

Stage 1 — Heat Cramps:

Painful muscle cramps, usually in the legs or abdomen. Heavy sweating. The person is still alert and able to communicate.

Action: Move to a cool, shaded place. Give water or a sports drink with electrolytes. Gently massage cramping muscles. Rest for several hours before any further activity in the heat.

Stage 2 — Heat Exhaustion:

Symptoms: Heavy sweating. Pale, cool, moist skin. Rapid but weak pulse. Nausea or vomiting. Dizziness or fainting. Headache. Weakness and fatigue. The person is still conscious and aware.

This is the critical intervention point. Treated promptly, heat exhaustion does not become heat stroke. Left untreated, it progresses rapidly.

Action — Step 1: Move the person to a cool, air-conditioned room immediately. If air conditioning is unavailable, place them in the shade and fan them vigorously.

Action — Step 2: Have the person lie down. Raise their feet about 30 centimetres (12 inches) above the level of their heart.

Action — Step 3: Remove excess clothing and any tight garments.

Action — Step 4: Apply cool, wet cloths to the skin — especially the forehead, back of the neck, armpits, and wrists. Fan the person to increase evaporative cooling.

Action — Step 5: Give cool water to drink — small sips, not large gulps. A sports drink with electrolytes is better if available. **Do not give alcohol or caffeine.**

Action — Step 6: Monitor the person closely. If symptoms do not improve within 30 minutes, or if they worsen (especially confusion or loss of consciousness), call emergency services immediately — this is now a heat stroke emergency.

Stage 3 — Heat Stroke:

Heat stroke occurs when the body's temperature regulation system fails completely and core body temperature rises above 40°C (104°F). This is a medical emergency.

Symptoms: Body temperature above 40°C. Confusion, agitation, slurred speech, or loss of consciousness. Hot, red skin that may be either dry OR wet. Rapid, strong pulse. Nausea and vomiting. Seizures in severe cases.

KEY DIFFERENCE: The most important sign of heat stroke vs heat exhaustion is confusion or change in mental state. If the person is confused, aggressive, unresponsive, or unconscious — this is heat stroke. Call emergency services immediately.

SECTION 2: TREATING HEAT STROKE — COOL FAST, CALL FAST

Heat stroke kills. Core body temperature must be reduced as rapidly as possible. Every minute of delay increases the risk of brain damage, organ failure, and death.

Step 1. Call emergency services (112) immediately. Do this first, then begin cooling.

Step 2. Move the person out of the heat and sun to the coolest available place — an air-conditioned room, a shaded area, or any cool interior.

Step 3. Remove outer clothing. Loosen all tight garments.

Step 4 — COOL BY THE BEST METHOD AVAILABLE:

Best method: Immerse the person in a bathtub of cold water, or use a cold shower. This is the fastest cooling method.

If no tub: Soak the person's clothing and skin with cold water continuously and fan vigorously. The combination of water and air movement is critical.

Apply ice packs or bags of frozen food to the neck, armpits, and groin — these areas have large blood vessels close to the skin and cool the blood efficiently.

Step 5. Do NOT give water or fluids by mouth if the person is confused, vomiting, or unconscious. There is a risk of choking.

Step 6. If the person is unconscious and not breathing, begin CPR.

Step 7. Continue cooling until emergency services arrive. Do not stop cooling to transport the person — cool first, transport second.

REMEMBER: Cool first, transport second. A person with heat stroke who is actively being cooled has far better survival odds than a person who is transported without cooling. Keep cooling throughout the journey to hospital.

SECTION 3: WHO IS MOST AT RISK

The following groups are most vulnerable during a heatwave and should be checked on twice daily:

- Elderly people (over 65), especially those living alone
- Infants and young children

- People with heart disease, kidney disease, diabetes, or high blood pressure
- People taking certain medications (diuretics, antihistamines, psychiatric medications) — these impair the body's ability to regulate heat
- People who work outdoors: farmers, construction workers, labourers
- People without access to air conditioning or cool shelter

SECTION 4: PROTECTING YOURSELF DURING A HEATWAVE

Water: Drink water continuously throughout the day, even before you feel thirsty. Thirst is a late sign of dehydration. Aim for at least 2 to 3 litres of water per day during a heatwave. Avoid alcohol and caffeine — they accelerate dehydration.

Timing: Stay indoors or in shade during the hottest hours of the day (11 AM to 4 PM in India). If you must be outdoors, take breaks in shade every 15 to 20 minutes.

Clothing: Wear loose, lightweight, light-coloured clothing. Light colours reflect sunlight. Loose clothing allows air to circulate around the skin.

Cooling your space: Close curtains and blinds on sun-facing windows during the day. Open windows at night when it is cooler to let cool air in. Wet a cloth and hang it near a fan — the evaporation cools the air.

Warning signs in yourself: Stopping sweating during extreme heat, feeling confused or dizzy, developing a headache that does not improve with water — these are signs to immediately move to a cooler place and seek help.

NEVER leave children, elderly people, or animals in a parked vehicle during hot weather. Car interiors can reach 50–60°C within minutes even on a moderately warm day. This is fatal.